



Bahir Dar University

Bahir Dar Institute of Technology

Faculty of Computing

DEPARTEMENT OF SOFTWARE ENGINEERING

Proposal submission

Name	ID
1. Bezawit Mamo	BDU1405561
2. Betelhem Tsega	BDU1404679
3. Marta Ejigu	BDU1404214

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Bahir Dar University, Bahir Dar Institute of Technology

Project Title 1: Smart Maternal & Child Health Follow-Up and Referral System

Introduction

Maternal and child health remains a major public health concern, particularly in developing regions where continuous monitoring during pregnancy and early childhood is limited. Many mothers lack access to reliable follow-up care, proper nutritional guidance, emergency support during labor (ምጥ), and structured child growth monitoring. Health centers also face challenges in tracking pregnancy progress, vaccination schedules, newborn registration, and high-risk referrals.

This project proposes the development of an AI-powered Maternal & Child Health Follow-Up and Smart Referral System designed to support mothers, healthcare providers, and local government (woreda) offices. The platform integrates AI-based pregnancy monitoring, baby nutrition recommendations, growth tracking, emergency ambulance requests, digital birth registration, and intelligent referral management into a single digital healthcare ecosystem. The system aims to improve accessibility, safety, and efficiency in maternal and child healthcare services.

Statement of the Problem

Current maternal and child healthcare systems suffer from fragmented monitoring, limited real-time communication, and delayed emergency response. Many mothers especially first-time mothers lack proper guidance on nutrition, hygiene, breastfeeding, and labor preparation. Child growth and vaccination tracking are often inconsistent, and newborn registration processes are manual and delayed. Additionally, high-risk pregnancies are not always identified early enough for timely referral to higher-level hospitals.

There is no unified digital platform that combines AI-based health guidance, growth monitoring, emergency support, and referral systems into one integrated solution. This gap results in preventable maternal and infant health complications. The proposed project addresses this challenge by creating a centralized, AI-supported follow-up and referral system that improves monitoring, accountability, and healthcare decision-making.

Objectives

The primary objective is to design and develop an AI-powered maternal and child health monitoring platform. Specific objectives include:

- Developing a pregnancy tracking system with weekly health updates
- Creating an AI-based baby nutrition and feeding recommendation system
- Enabling real-time child growth and development monitoring
- Integrating vaccination tracking and reminder notifications
- Allowing mothers to request ambulance services during labor (ምጥ)
- Implementing digital newborn registration linked to woreda systems

- Designing a smart referral system for high-risk pregnancy and delivery cases
- Providing educational resources for first-time mothers
- Ensuring data privacy and secure communication

Significance of the Project

This project improves maternal and child health outcomes by enabling early risk detection, continuous monitoring, and faster emergency response. It increases healthcare accessibility, strengthens vaccination compliance, and improves accountability in birth registration systems.

Technologically, the system demonstrates how AI and digital health solutions can modernize maternal healthcare infrastructure. Socially, it contributes to reducing maternal and infant mortality rates while supporting informed, confident motherhood. The platform also provides a scalable model that can be implemented across multiple health centers and regions.

Project Title 2: Development of an Intelligent Resume Screening and Candidate Ranking System

Introduction

Recruitment and hiring in organizations can be slow, inconsistent, and biased due to manual resume evaluation. Human recruiters often face high workloads, making it difficult to screen large numbers of applicants efficiently. Moreover, evaluating candidates based on objective skill alignment, experience, and potential is challenging. This project proposes an AI-powered Resume Screening System that automatically analyzes, scores, and ranks candidates based on their CVs. The system streamlines the hiring process, reduces human bias, and ensures that employers identify the most qualified candidates quickly and fairly.

Statement of the Problem

Organizations face several recruitment challenges:

- High volume of applications: Reviewing hundreds or thousands of CVs manually is time-consuming.
- Inconsistent evaluation: Different recruiters may rate the same candidate differently.
- Difficulty matching skills to job requirements: Recruiters may overlook qualified candidates due to lack of systematic evaluation.
- Bias in hiring decisions: Manual screening may unintentionally favor certain candidates over others.

These issues lead to delayed hiring, suboptimal candidate selection, and potential loss of talent.

Objectives

The goal of this project is to develop an AI-driven platform that streamlines candidate selection and ensures objective evaluation:

- **CV Upload & Parsing:** Allow candidates to upload resumes, which the system parses for education, experience, skills, and certifications.
- **Machine Learning Candidate Ranking:** Implement AI/ML algorithms to score and rank candidates based on relevance to job requirements.
- **Skill & Experience Matching:** Automatically compare candidate qualifications against job criteria for objective evaluation.
- **Dashboard & Analytics for Recruiters:** Provide a user-friendly interface showing top-ranked candidates, skill gaps, and comparative scores.

Significance of the Project

- Saves time and effort for HR and recruitment teams.
- Provides objective, data-driven candidate evaluation to reduce bias.
- Helps organizations identify the most qualified candidates quickly.
- Introduces students to AI applications in industry-relevant contexts.
- Scalable to small businesses, NGOs, and large companies across Ethiopia.